ECE264D:
CMOS Analog Integrated Circuits and Systems IV

Lecture 5:
Integer-N Frequency Synthesis
Frequency Dividers

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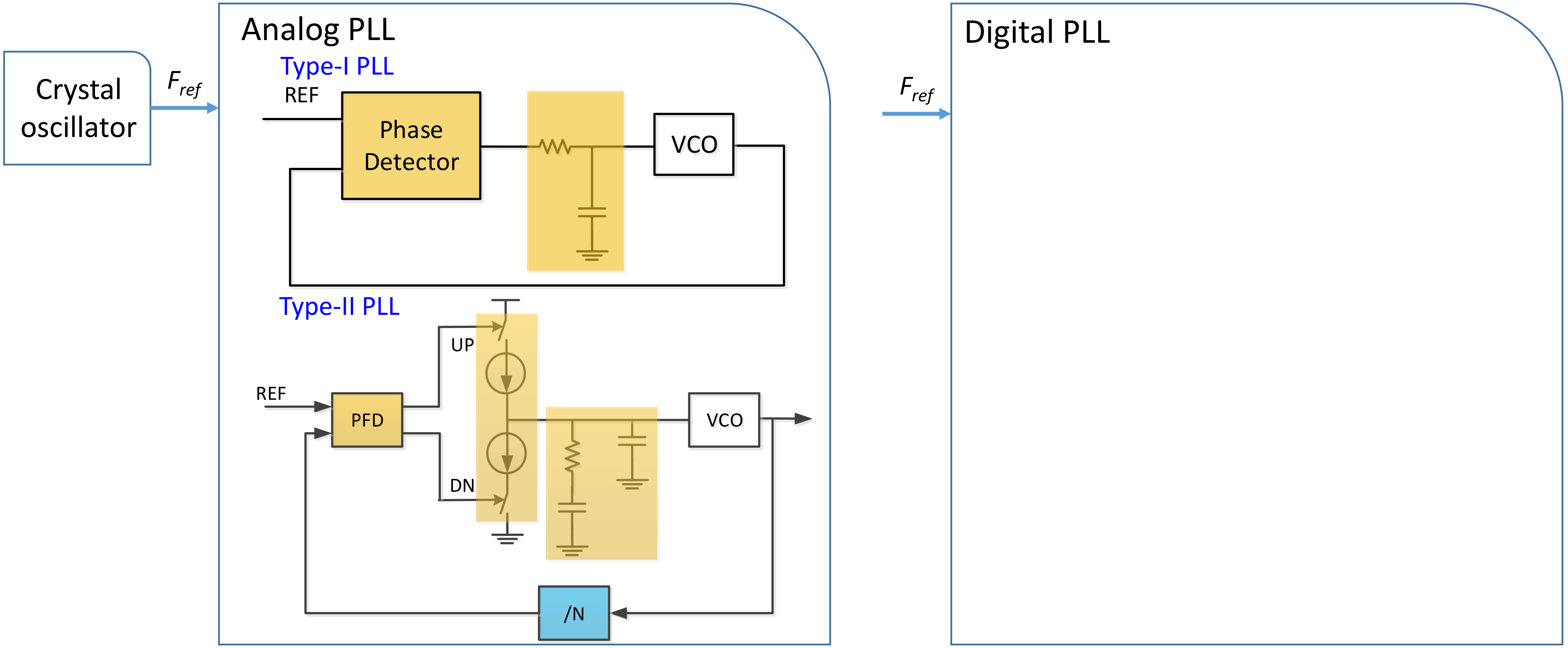
Frequency Synthesizer Overview

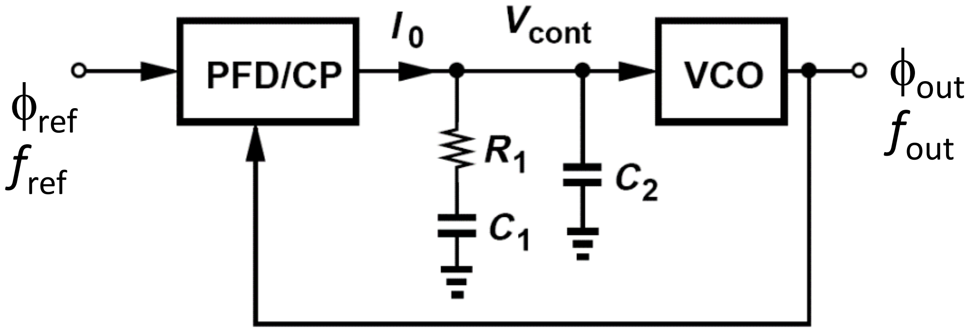


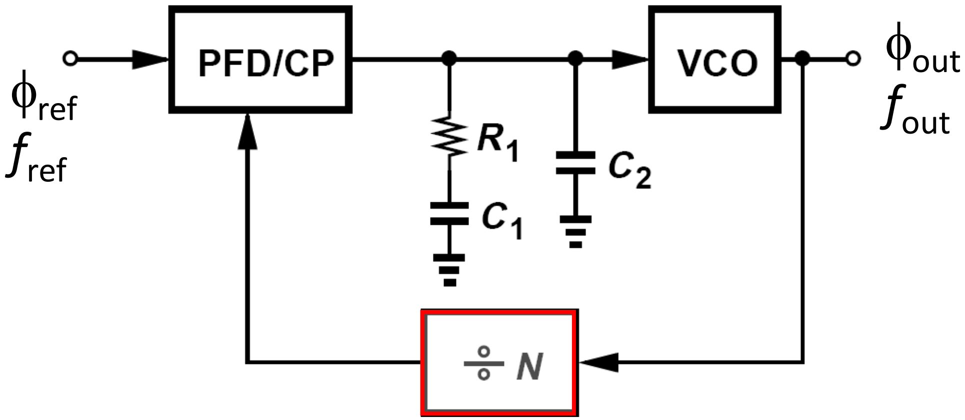
Outline

- Frequency dividers and frequency synthesis
- Frequency divider building blocks
  - D flip flops
    - Static master-slave DFF
    - TSPC (true single-phase clock) DFF
    - CML (current mode logic) DFF
  - Divide-by-two divider
    - Johnson counter
    - Razavi divider
    - Miller divider
    - Injection locked frequency divider
  - Prescalers
  - Counters
  - Pulse-swallow divider

Note:
Dividers for frequency synthesis
vs.
Dividers for LO generation

Frequency Synthesis by Phase Lock





After locking, $\phi_{out}=\phi_{ref}$, $f_{out}=f_{ref}$

After locking, $\phi_{out}=\phi_{ref}$, $f_{out}=N \cdot f_{ref}$

- By adding a frequency divider div-N in the feedback path, the output frequency equals to N times of the input frequency.
- By changing N ratio, f_{out} can be programmed to different value ➔ Frequency synthesis
- Frequency programmability resolution is $f_{ref.}$
- In practice, N ranges from a single digit to a few hundred or more.

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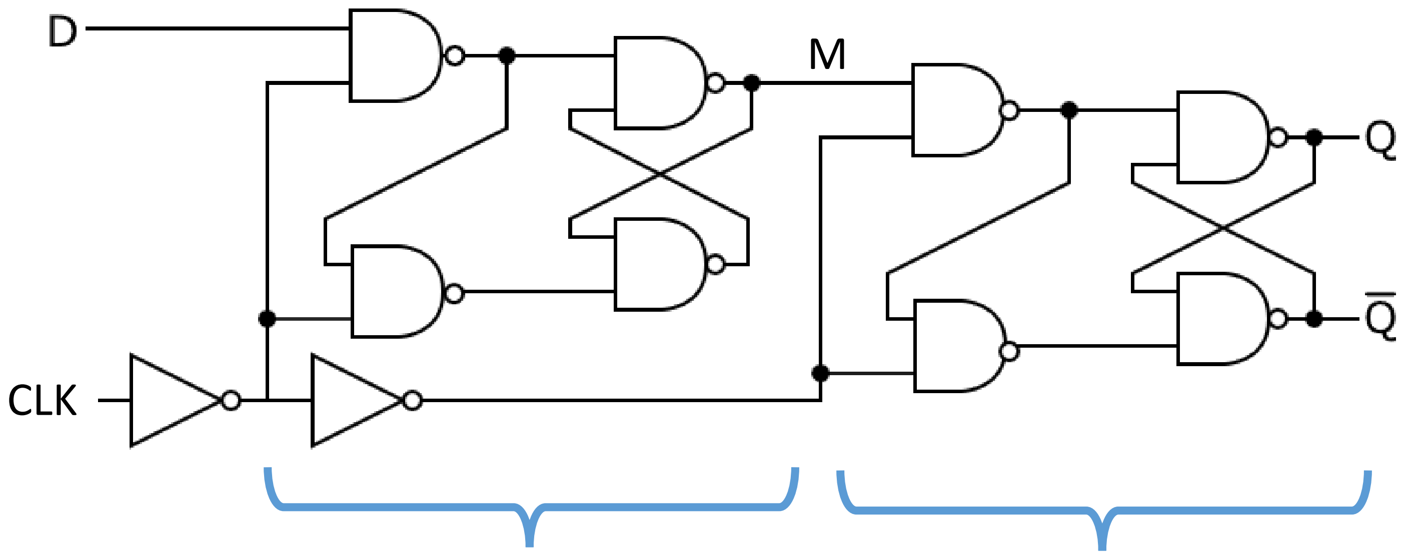
CMOS Static Master Slave DFF – NAND Gate Based



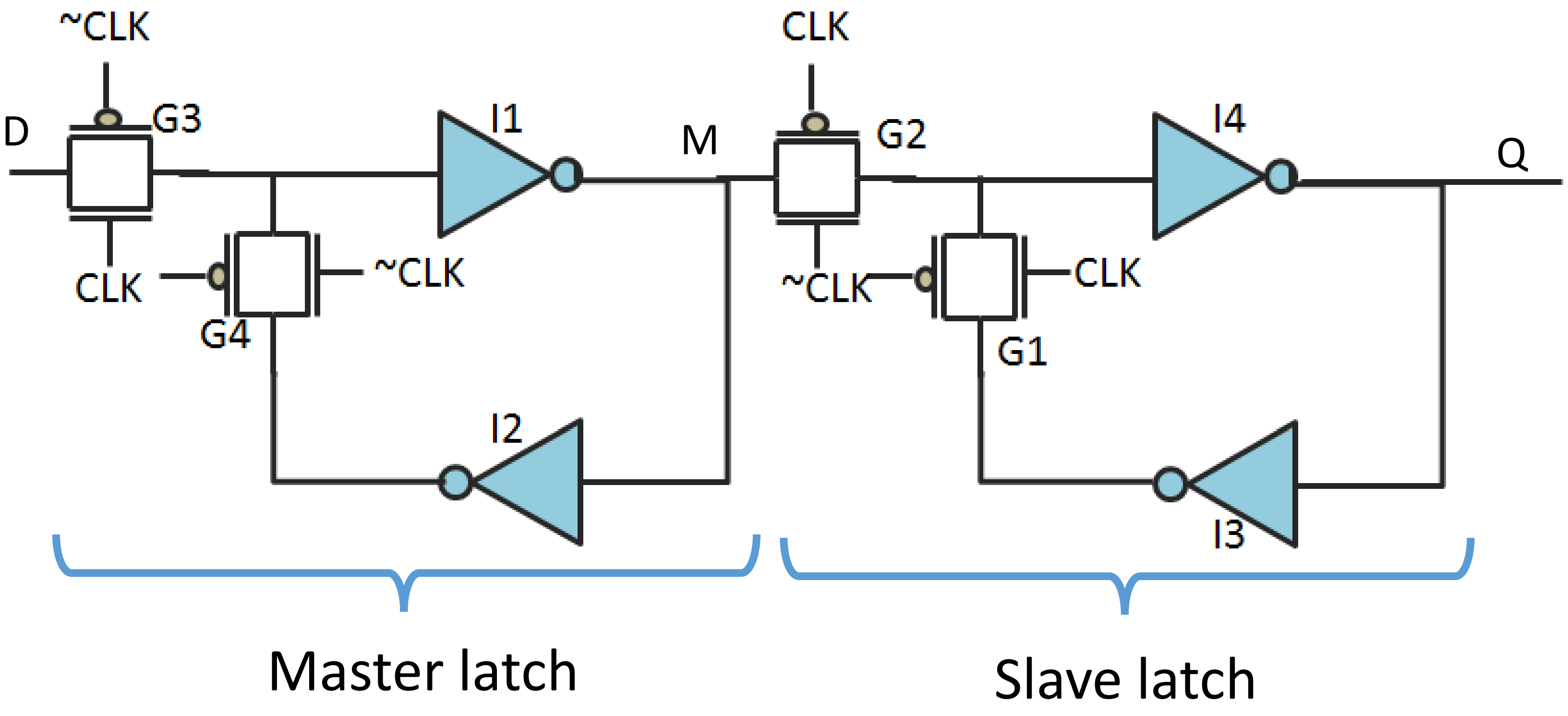
Master latch

Slave latch

Clock	D	Q_{next}
Rising edge	0	0
Rising edge	1	1
Non-Rising	X	Q

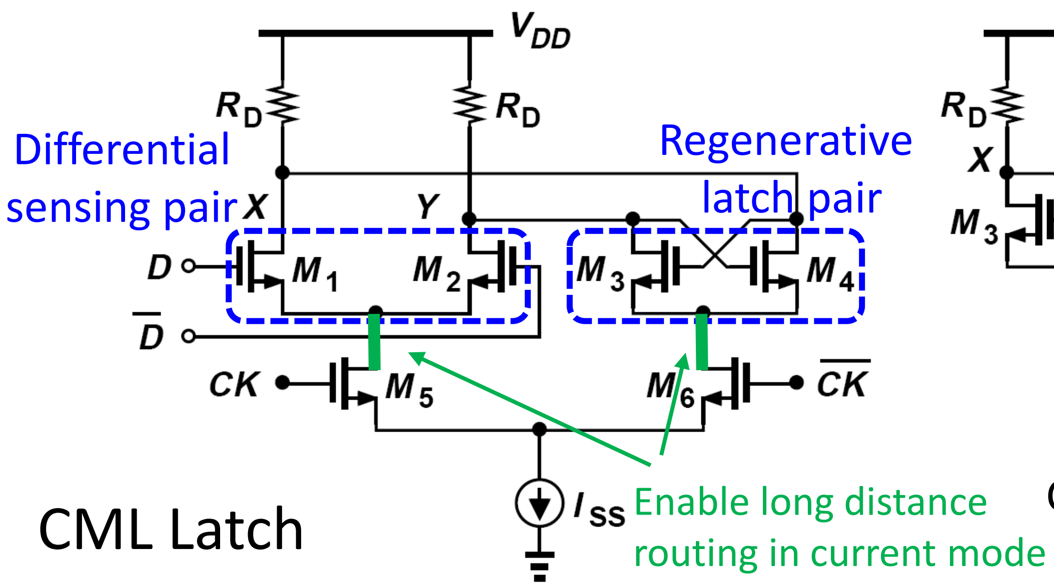
- Classical DFF with master latch and slave latch
- Static circuit, no minimum operating frequency constraint
- Master latch output M updates when CLK=0 and freezes when CLK=1
- Slave latch output Q updates when CLK=1 and freezes when CLK =0

CMOS Static Master Slave DFF – Transmission Gate Based

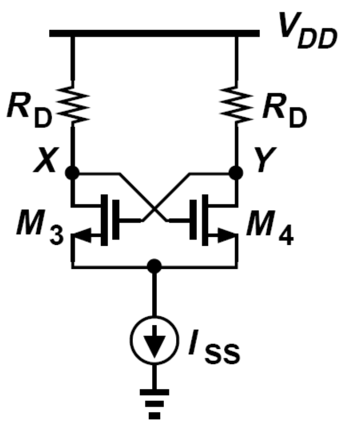


- Edge triggered DFF with a master latch and a slave latch
- Static circuit, no minimum operating frequency constraint
- Master latch output M updates when CLK=1 and freezes when CLK=0
- Slave latch output Q updates when CLK=0 and freezes when CLK =1

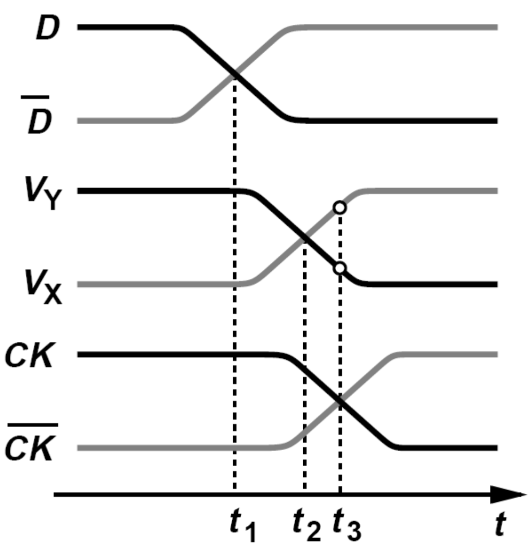
Current Mode Logic (CML) DFF

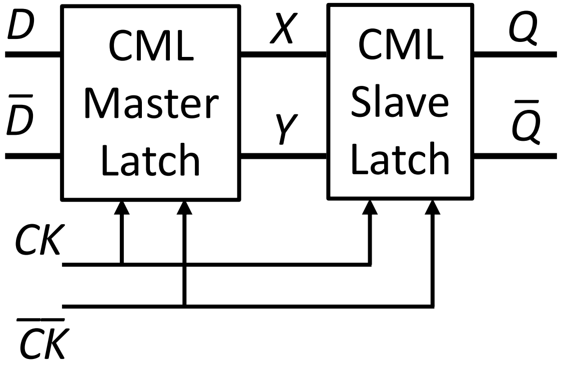


CML Latch



CK=0





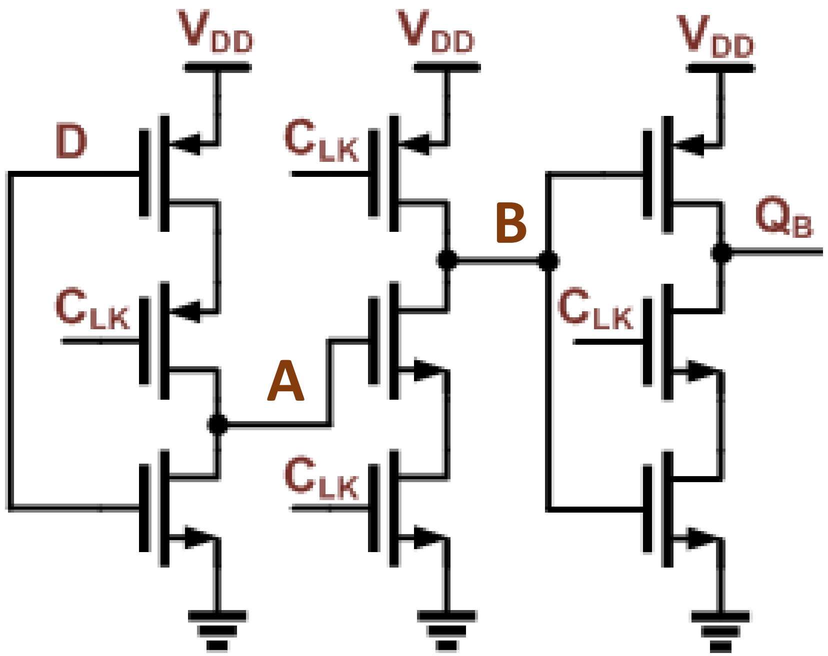
CML DFF

- CMOS based DFF has multi-gate delay that limits maximum speed
- Differential pair in CML can be quickly enabled/disabled through tail current source ⇨ fast speed
- CK=1, sensing mode: M1 /2 sense and amplify input D, V_X/V_Y track input
- CK=0, regenerative mode, M3/4 retain V_X/V_Y

☺ Fast speed
☺ Generate less VDD spur
☺ Generate differential output
☺ Feasible to route long distance from CK input to DFF core
☹ Constant current I_{ss}
☹ Require different input and differential clock
☹ Small swing, less headroom

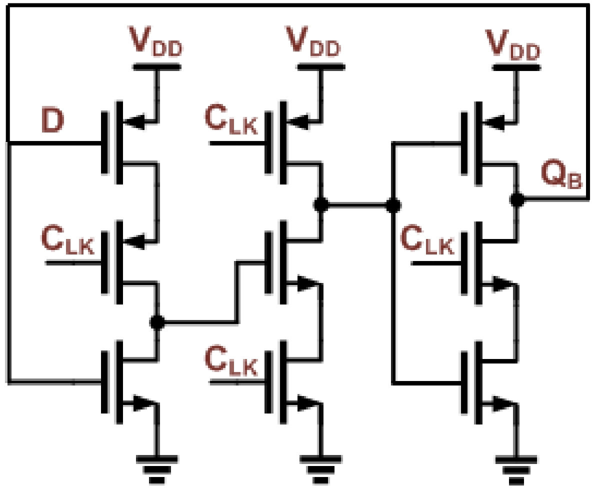
Ref [1]

True Single Phase Clock (TSPC) DFF



- ☺ Less transistors than conventional CMOS DFF; fast speed and smaller area
- ☺ Requires only single-ended CLK
- ☺ No static current; dynamic current depending on clock frequency
- ☺ Rail to rail swing
- ☹ VDD current spurs
- ☹ Dynamic circuit, F_{clk} has a minimum operating frequency (why? Which PVT condition prone to fail at low F_{clk}?

- CK=0: A tracks input D ($A = \bar{D}$), B is precharged to 1
- CK 0 → 1:
  - A retains its value at CK rising edge
  - B stays high if A=0 or B is discharged if A=1 ($B = \bar{A}$)
  - $Q_B = \bar{B}$



TSPC DFF based DIV-by-2

Ref [3]

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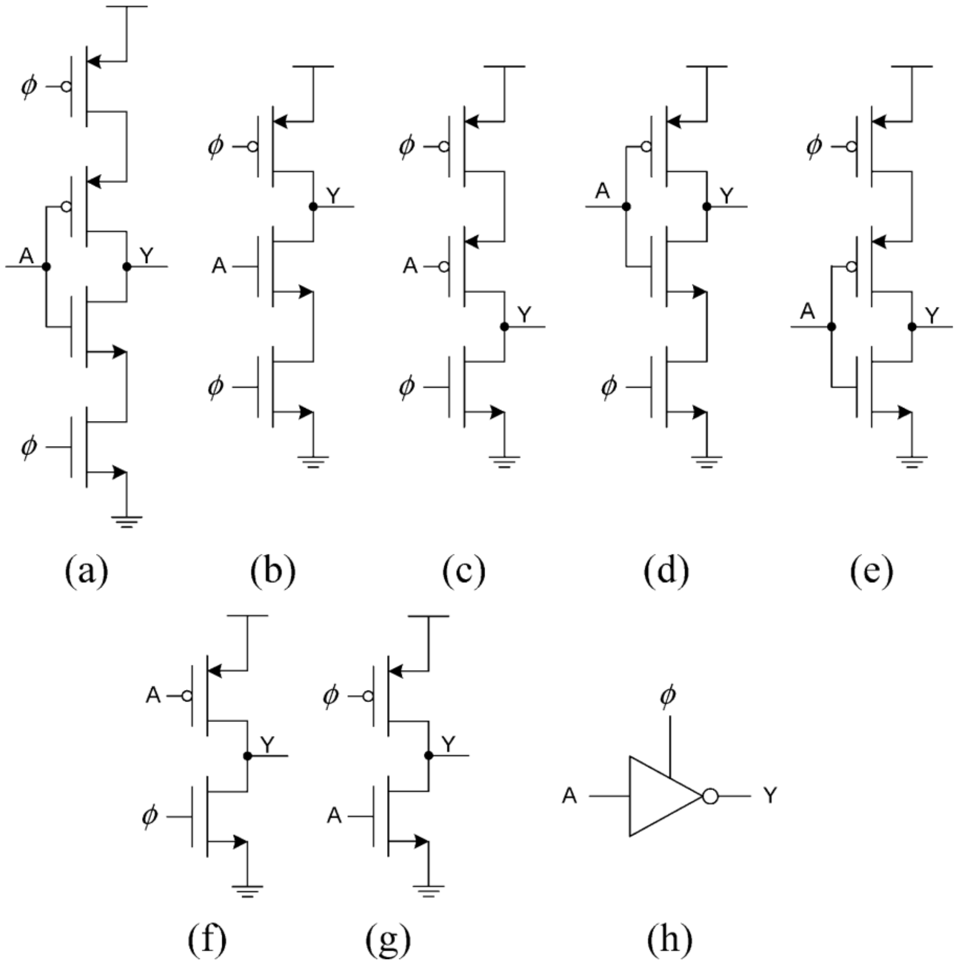
True Single Phase Clock (TSPC) DFF

Different building blocks of TSPC stage



A: signal input
ϕ: clock input

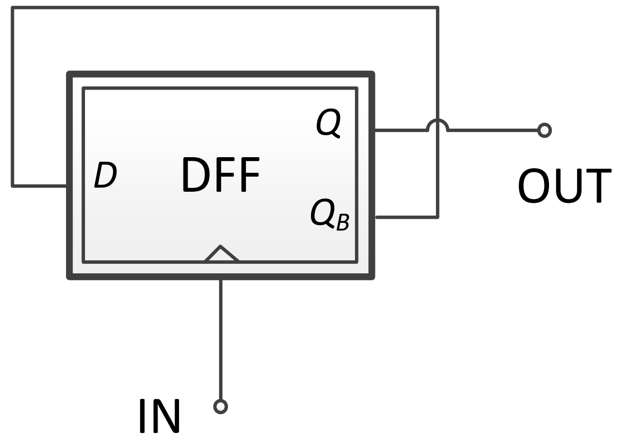
P: PMOS
N: NMOS
C: complementary

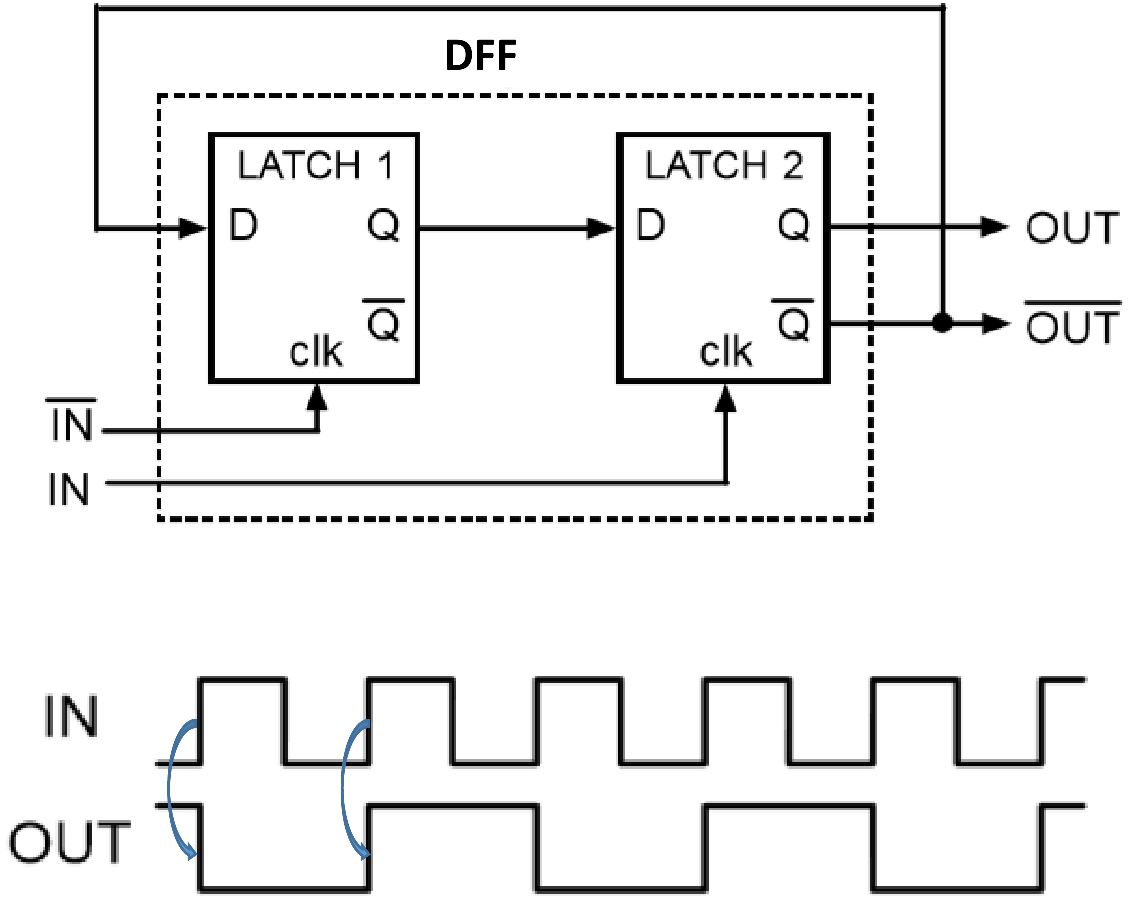
	Logic	Type	Input: A	Ouput: Y $\phi = 1$	Ouput: Y $\phi = 0$
(a)	CC	ratioless	1	0	z
			0	z	1
(b)	CN	ratioless	1	0	1
			0	z	1
(c)	CP	ratioless	1	0	z
			0	0	1
(d)	NC	ratioless	1	0	z
			0	1	1
(e)	PC	ratioless	1	0	0
			0	z	1
(f)	NP	ratioed	1	0	z
			0	r	1
(g)	PN	ratioed	1	0	r
			0	z	1

Clock Signal

Ref [7]

Divide-by-2 Divider (Johnson Counter)

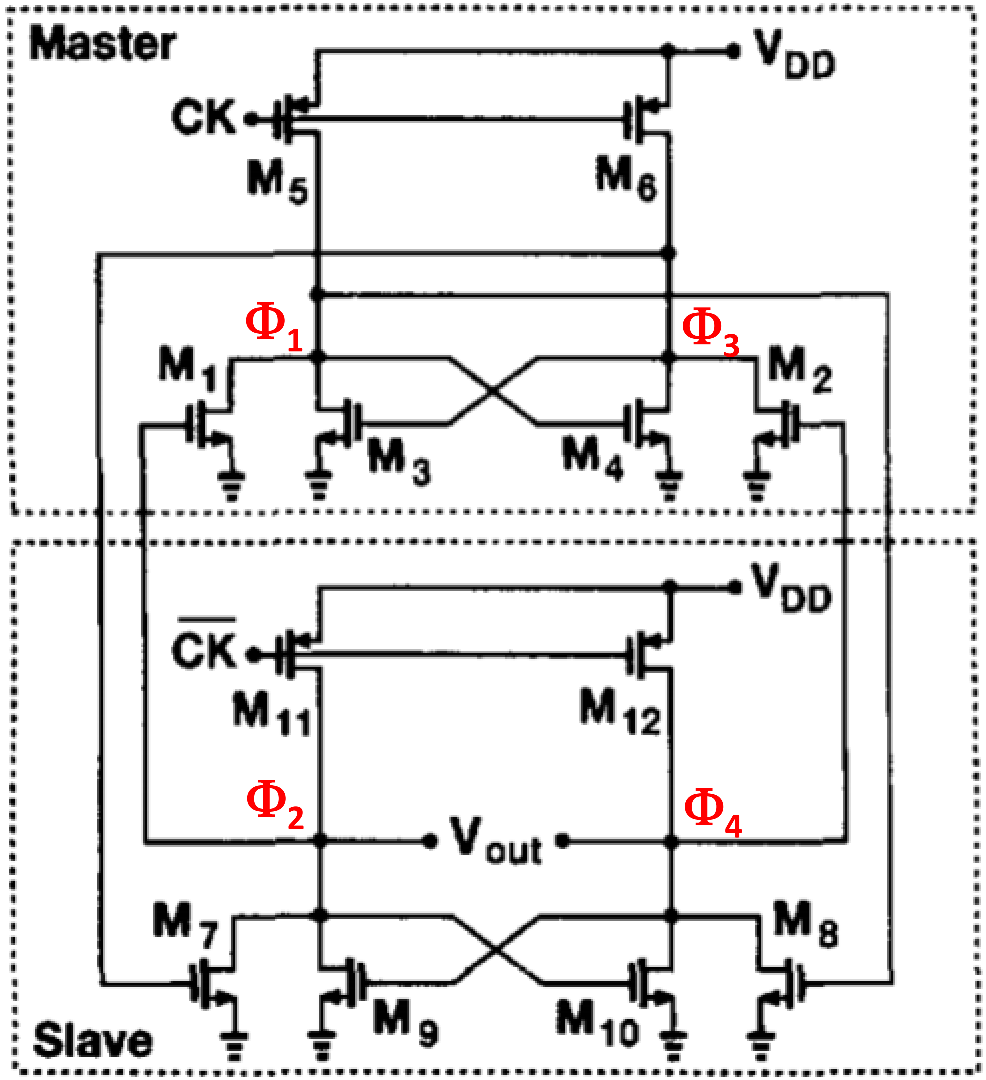




- DFF in a negative feedback loop
- Input as clock, output from Q
- The state machine has 2 states
- DFF can be any type (static M/S, CML, TSPC, etc)

D (QB)	Q
0	1
1	0

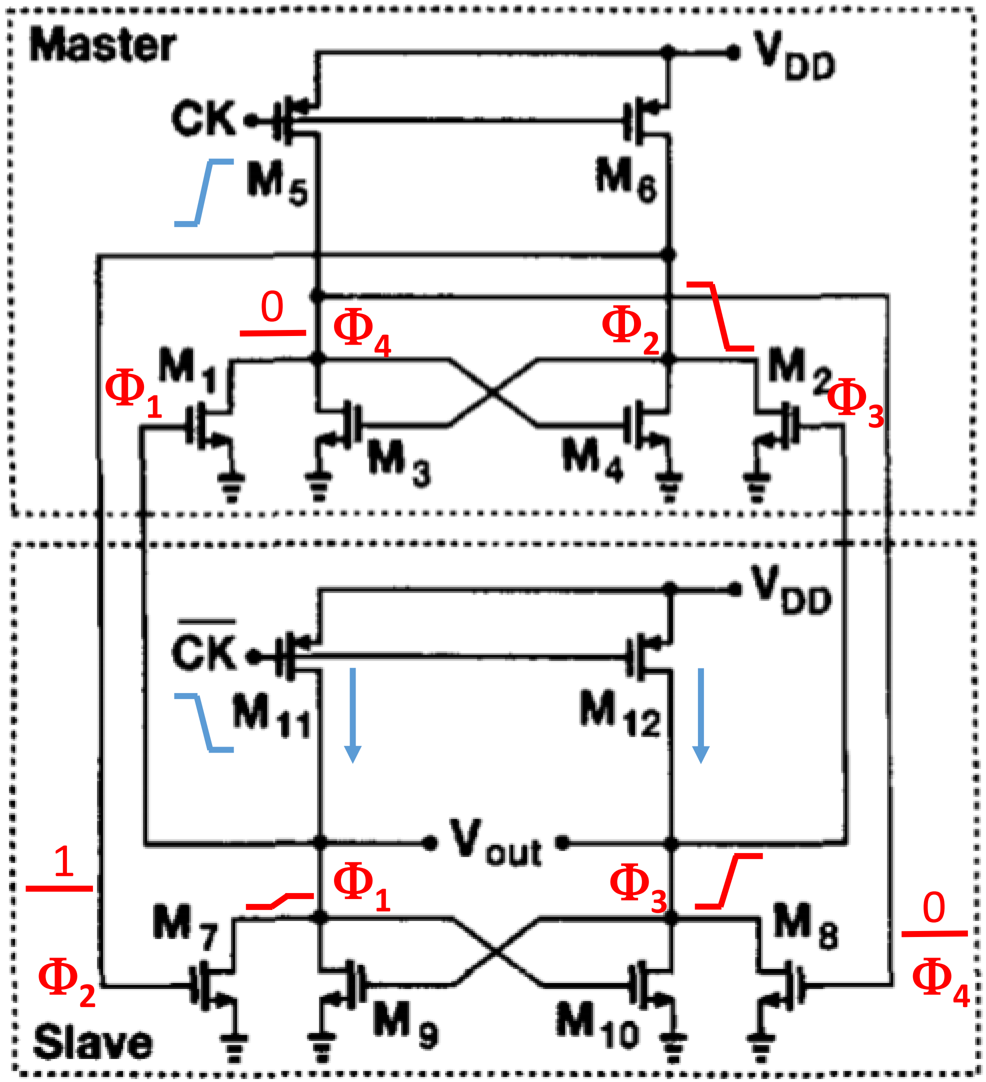
Razavi Divider

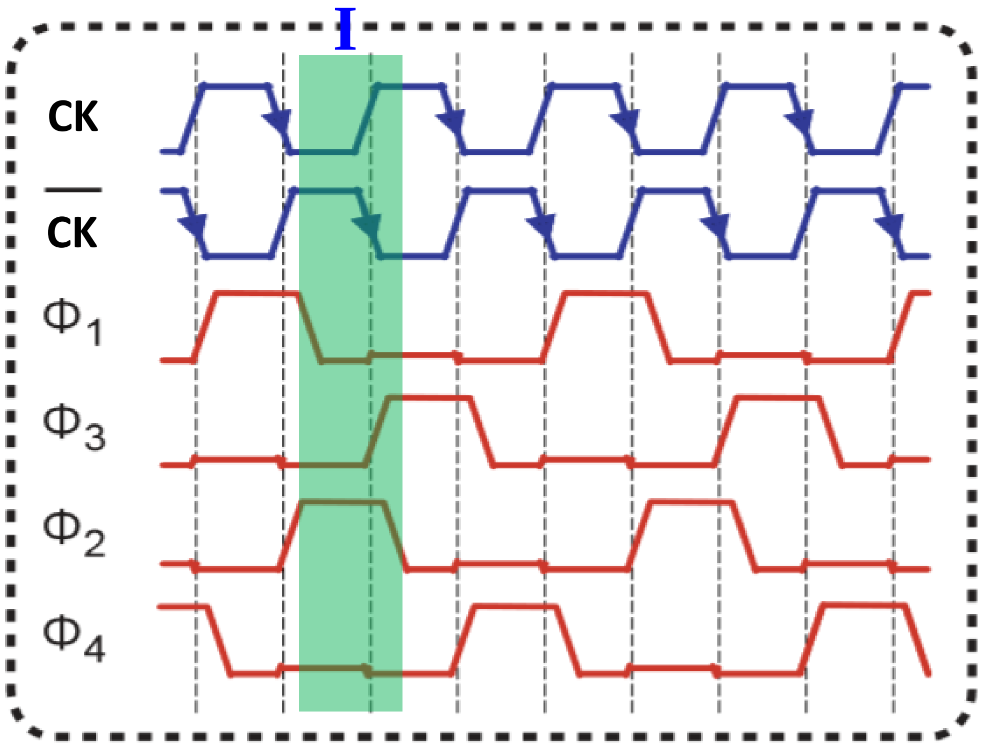


- Two latches: master latch, slave latch
- Each latch has
  - Two sense devices (M1/2 in master latch, M7/8 in slave latch)
  - Two regenerative devices (M3/4 in master latch, M9/10 in slave latch)
  - Two pull-up devices (M5/6 in master latch, M11/12 in slave latch)
- Two modes of operation
  - CK=high, M5/6 off, M11/12 on: master latch in sense mode, slave latch in store mode
  - CK=low, M5/6 on, M11/12 off: master latch in store mode, slave latch in sense mode

Ref [5]

Razavi Divider

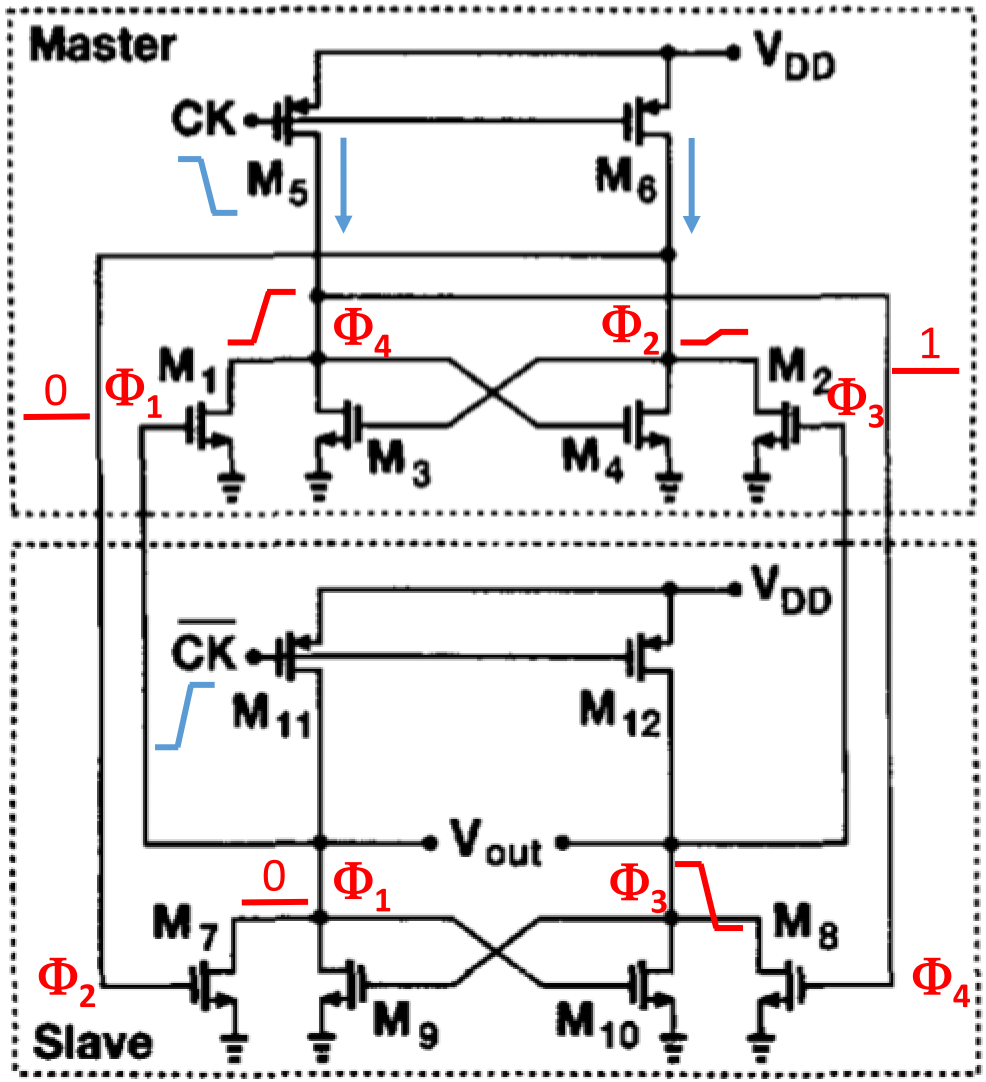


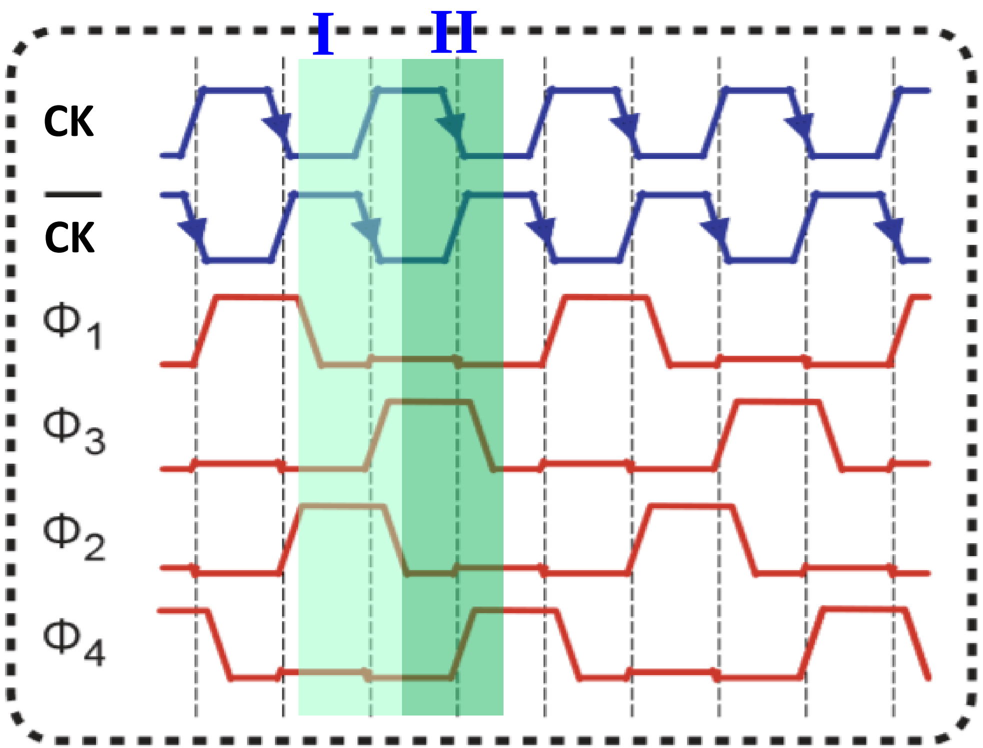


- Initial state: Φ_1=0, Φ_3=0, Φ_2=1, Φ_4=0
- @CKb, Φ_2=1 ➡ Φ_1 remains 0; Φ_4=0 ➡ Φ_3 0→1
- Φ_3 0→1 ➡ Φ_2 1→0; Φ_1=0, CK=1 ➡ Φ_4 remains 0

Ref [5]

Razavi Divider

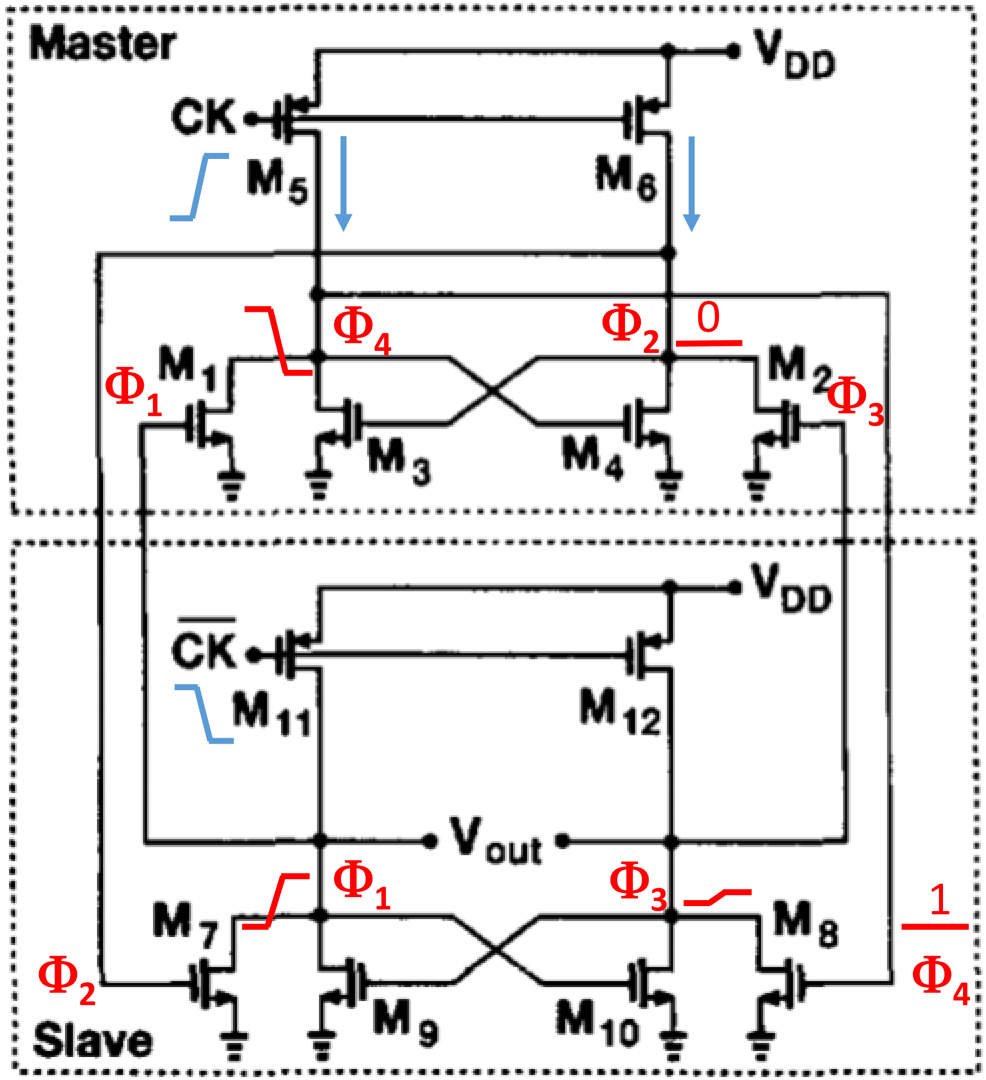


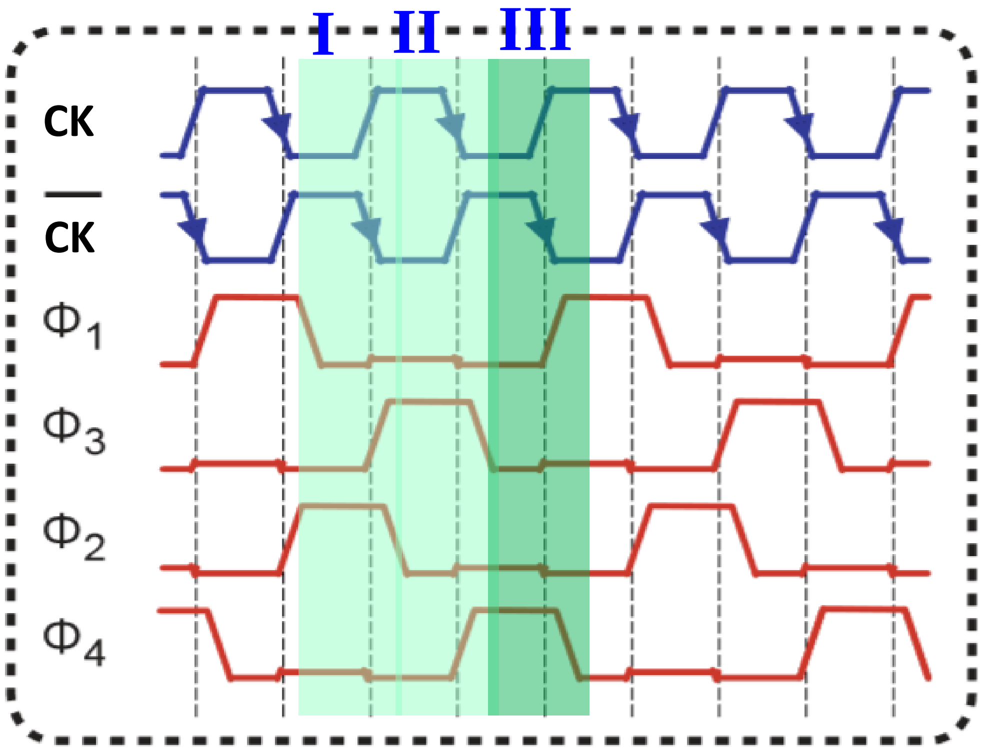


- @ end of state I: Φ_1=0, Φ_3=1, Φ_2=0, Φ_4=0
- @ CK, Φ_3=1 ➡ Φ_2 remains 0; Φ_1=0 ➡ Φ_4 0→1
- Φ_4 0→1 ➡ Φ_3 1→0; Φ_2=0, CKb=1 ➡ Φ_1 remains 0

Ref [5]

Razavi Divider

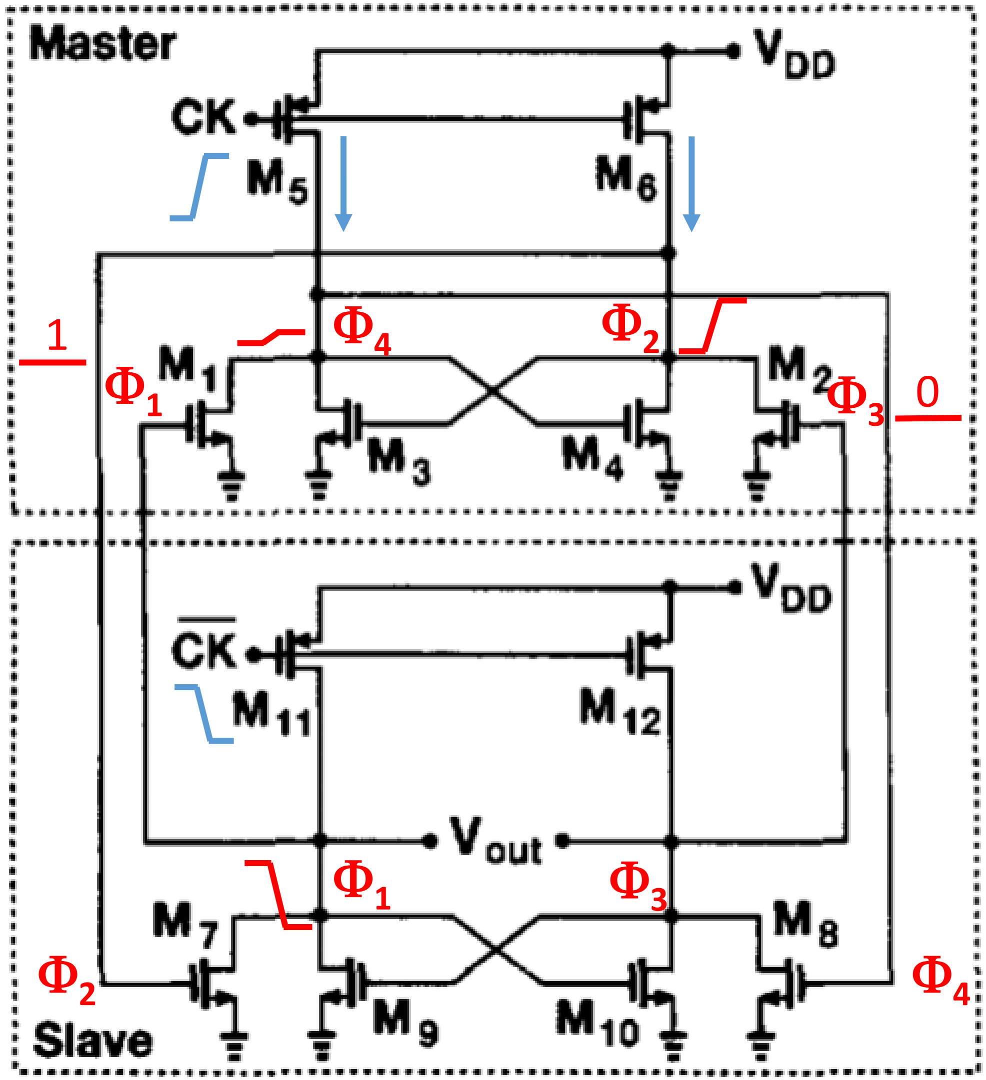


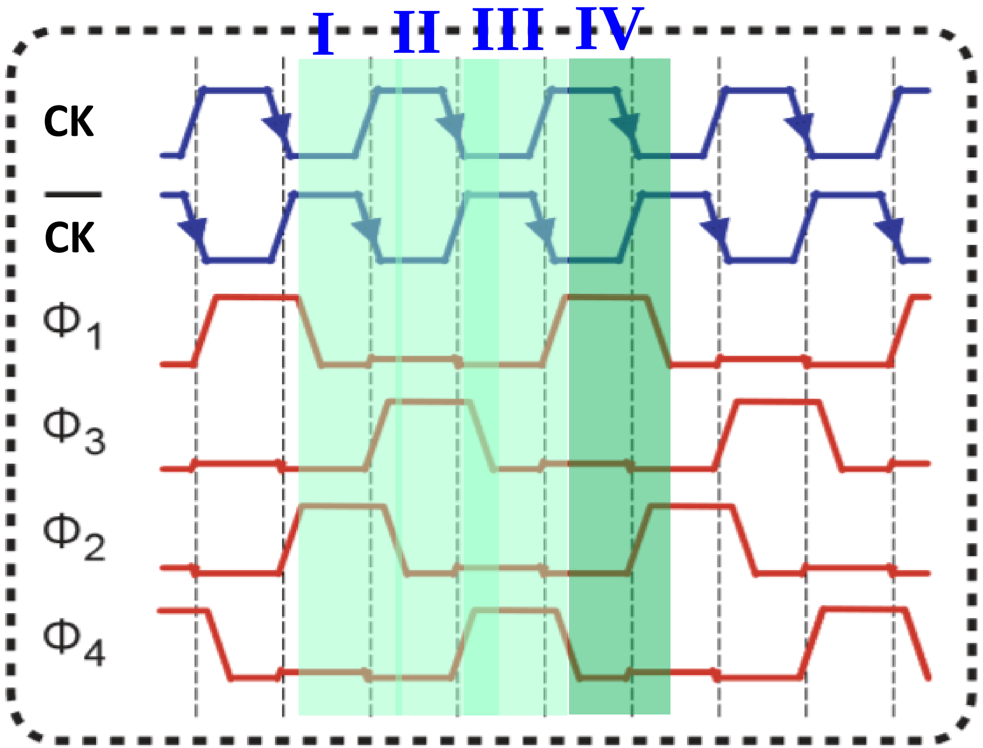


- @ end of state II: Φ_1=0, Φ_3=0, Φ_2=0, Φ_4=1
- @ CKb, Φ_4=1 ➡ Φ_3 remains 0; Φ_2=0 ➡ Φ_1 0→1
- Φ_1 0→1 ➡ Φ_4 1→0; Φ_3=0, CK=1 ➡ Φ_2 remains 0

Ref [5]

Razavi Divider





- @ end of state III: Φ_1=1, Φ_3=0, Φ_2=0, Φ_4=0
- @ CK ⎺_, Φ_1=1 ➡ Φ_4 remains 0; Φ_3=0 ➡ Φ_2 0→1
- Φ_2 0→1 ➡ Φ_1 1→0; Φ_4=0, CKb=1 ➡ Φ_3 remains 0

Ref [5]

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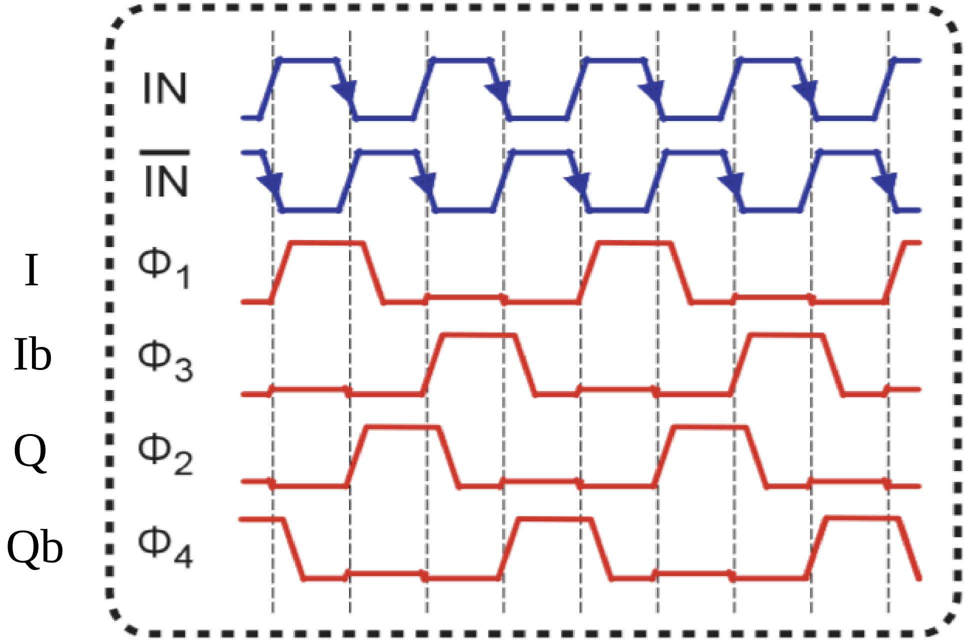
Razavi Divider



Φ_1	Φ_2	Φ_3	Φ_4
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

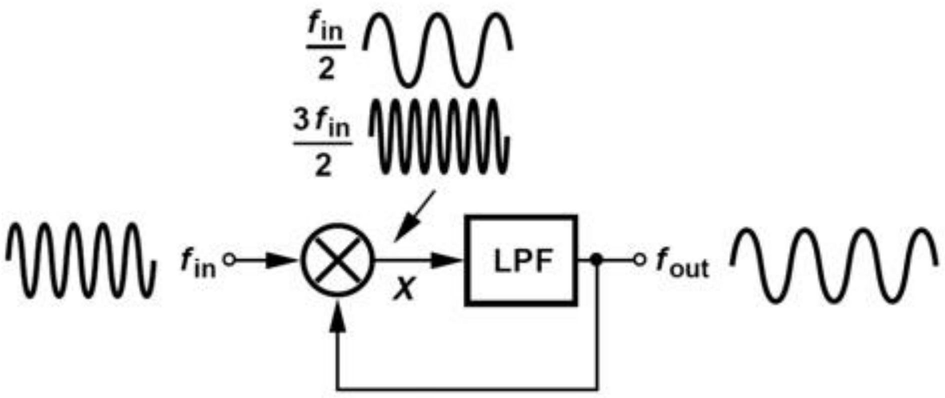
- Four states to complete one output period in two input clock cycles → Div 2
- Four outputs each has 25% duty cycle → non-overlapping outputs
  - Useful for quadrature LO signal generation for passive mixers: I, Ib, Q, Qb
- Full swing differential input clock (usually from VCO or VCO buffer)

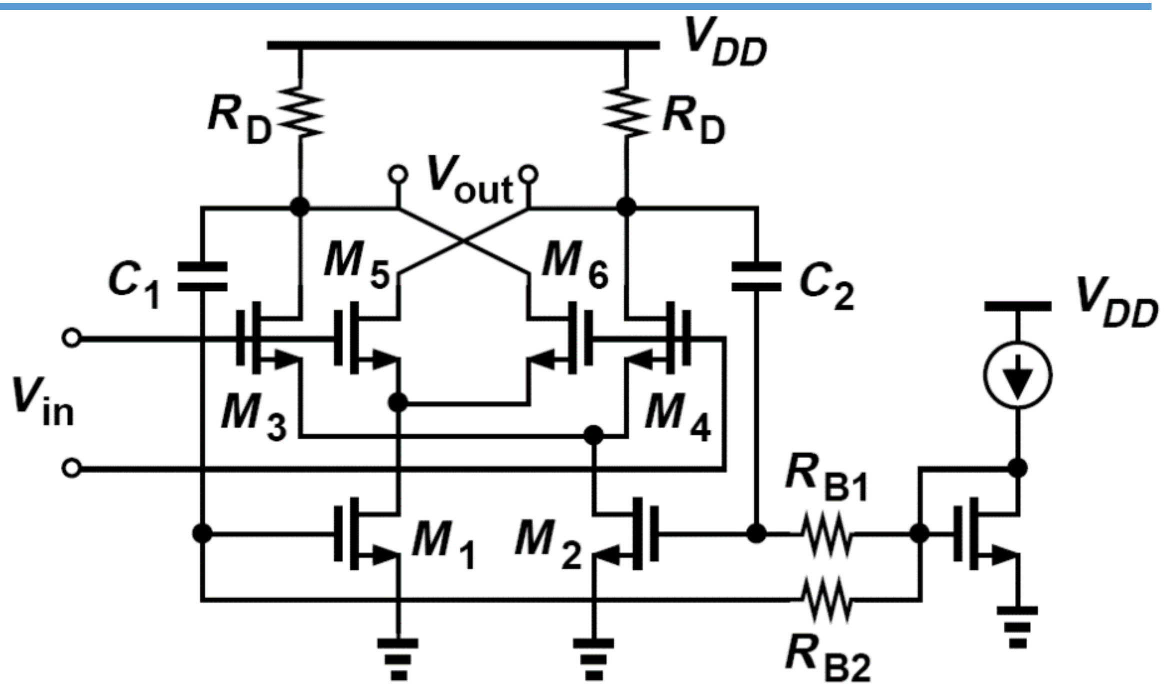
☺ Higher speed: no stacked PMOS; signal goes through only two gates per cycle

☺ Rail to rail swing, 25% duty cycle

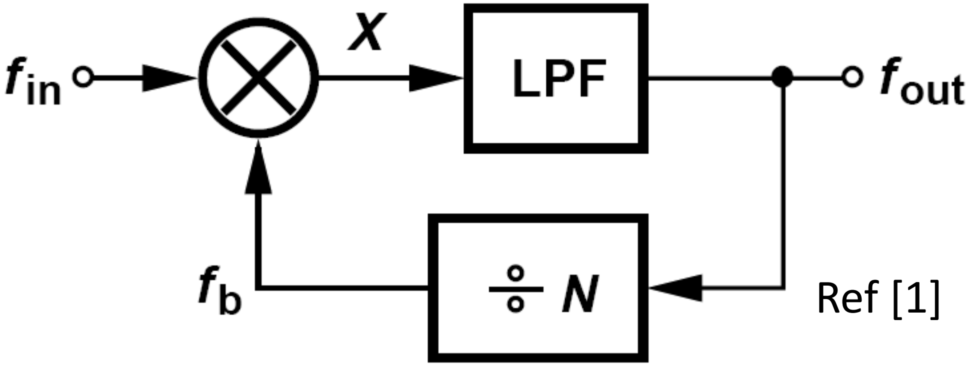
☹ Static current (e.g. current thru M11, M9)

Miller Divider





- $f_{x1} = f_{in} + f_{out}$, $f_{x2} = f_{in} - f_{out}$
- Only f_{x2} passes through LPF, f_{out}=f_{x2}
- $f_{out} = \frac{1}{2} f_{in}$
- With Div-N in feedback path, $f_{out} = \frac{N}{N+1} f_{in}$
- V_{in} at mixer LO input port, V_{out} returns to RF input port.



Ref [1]

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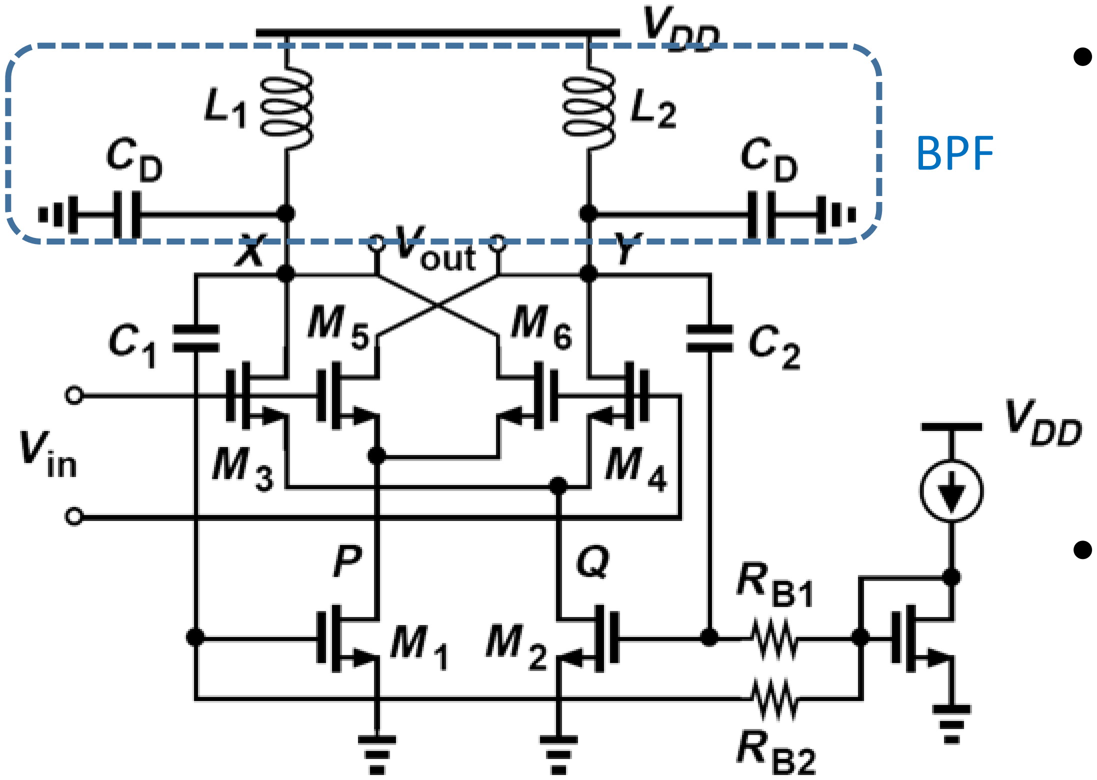
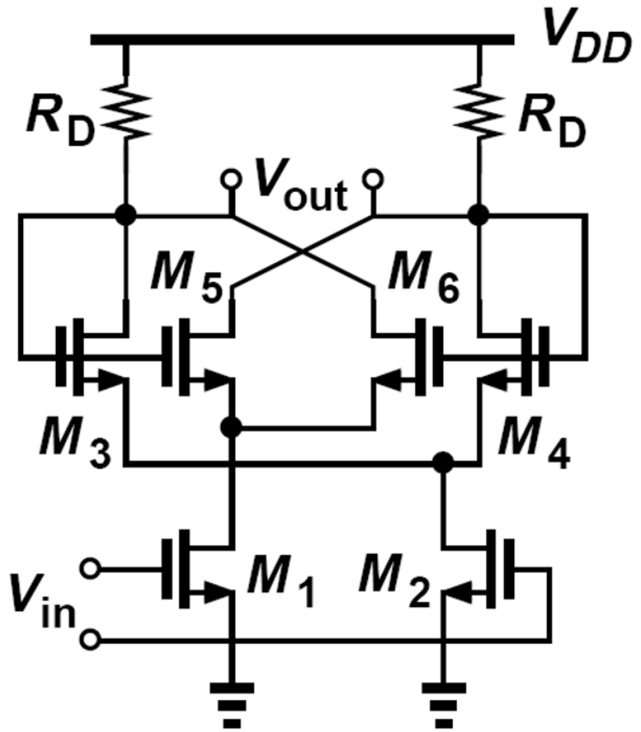
Miller Divider

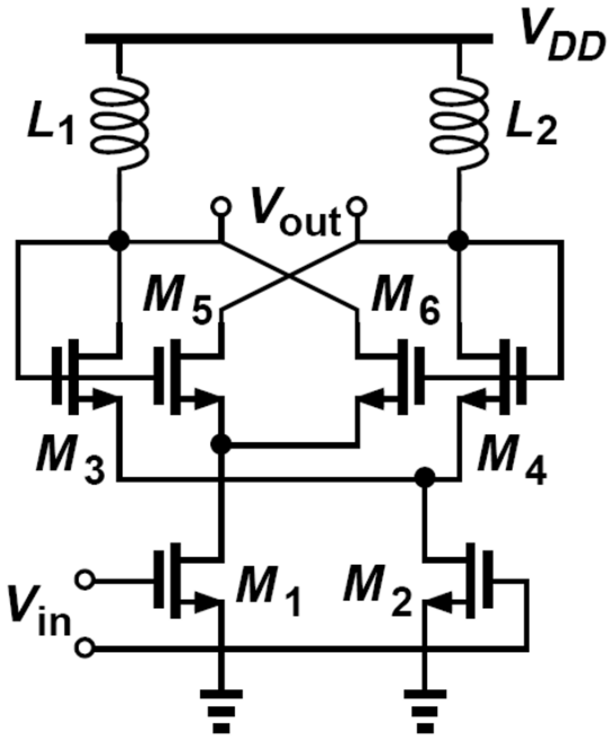


- To divide properly
  - Loop gain at $f_{in}/2 > 1$ ⟹ set max operation frequency
  - Loop gain at $3f_{in}/2 << 1$ ⟹ set min operation frequency
  - Dynamic divider
- High speed because the circuit does not rely on latching but rather on high frequency loop gain.

Ref [1]

Miller Divider





- Alternative Miller divider configurations
  - V_{in} at the RF input, V_{out} feedback to LO input
  - Resistive or inductive loading
- M5 and M6 cross coupled pair becomes positive feedback

Ref [1]

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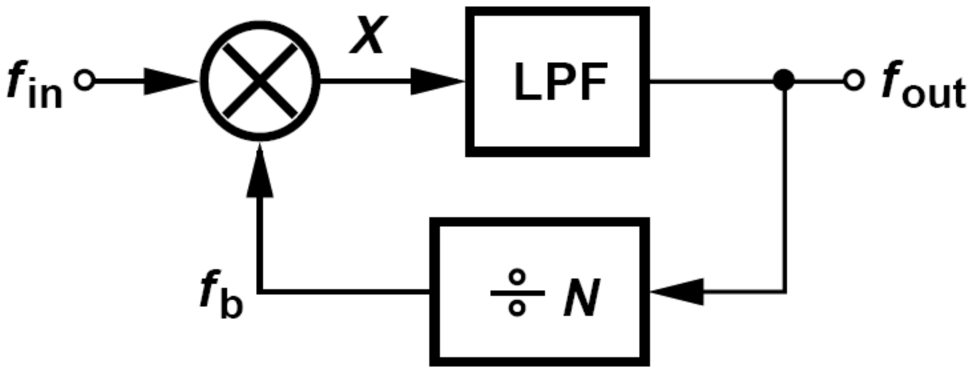
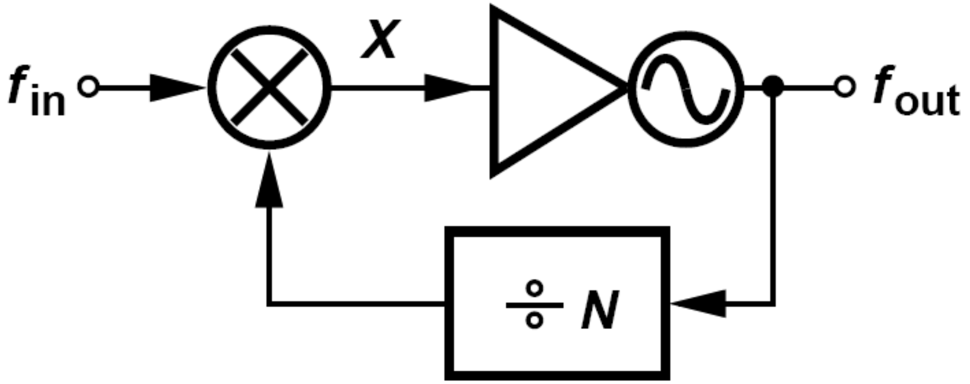
Injection Locked Frequency Divider (ILFD)





	Miller Divider	Injection Locked Divider
f_{out}	$\frac{N}{N+1} f_{in}$	$\frac{N}{N+1} f_{in}$ or $\frac{N}{N-1} f_{in}$
Require self oscillation w/o f_{in}	No	Yes
Operation frequency range	Wide	Narrow

Ref [1]

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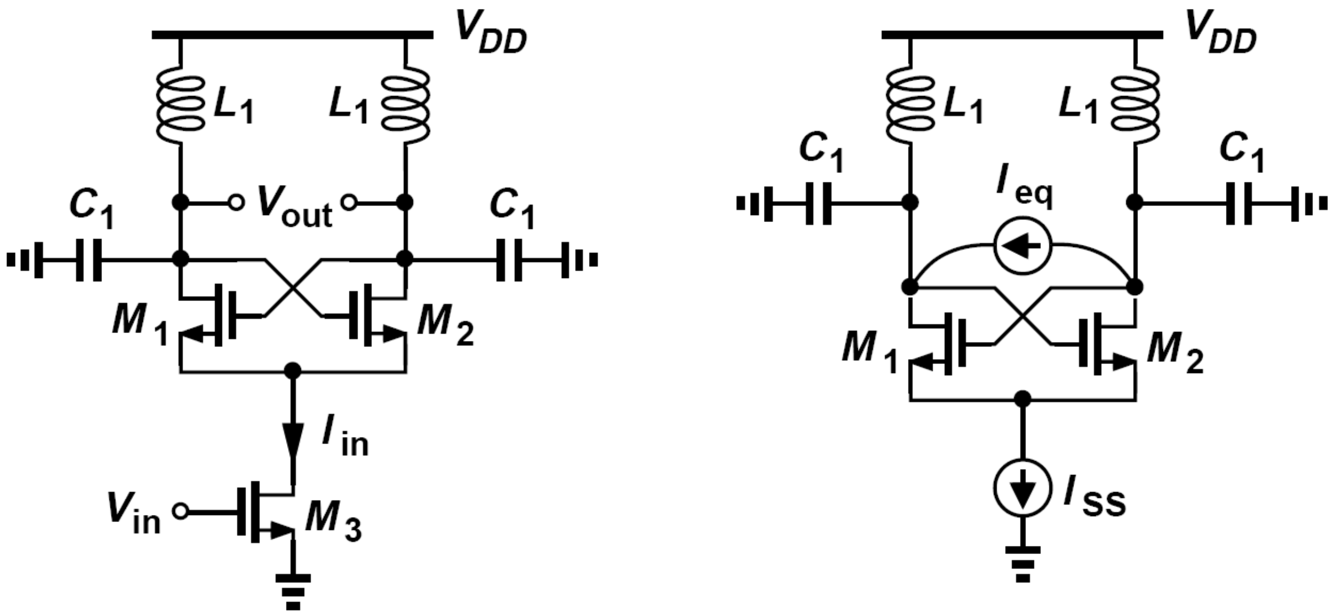
Injection Locked Frequency Divider (ILFD)

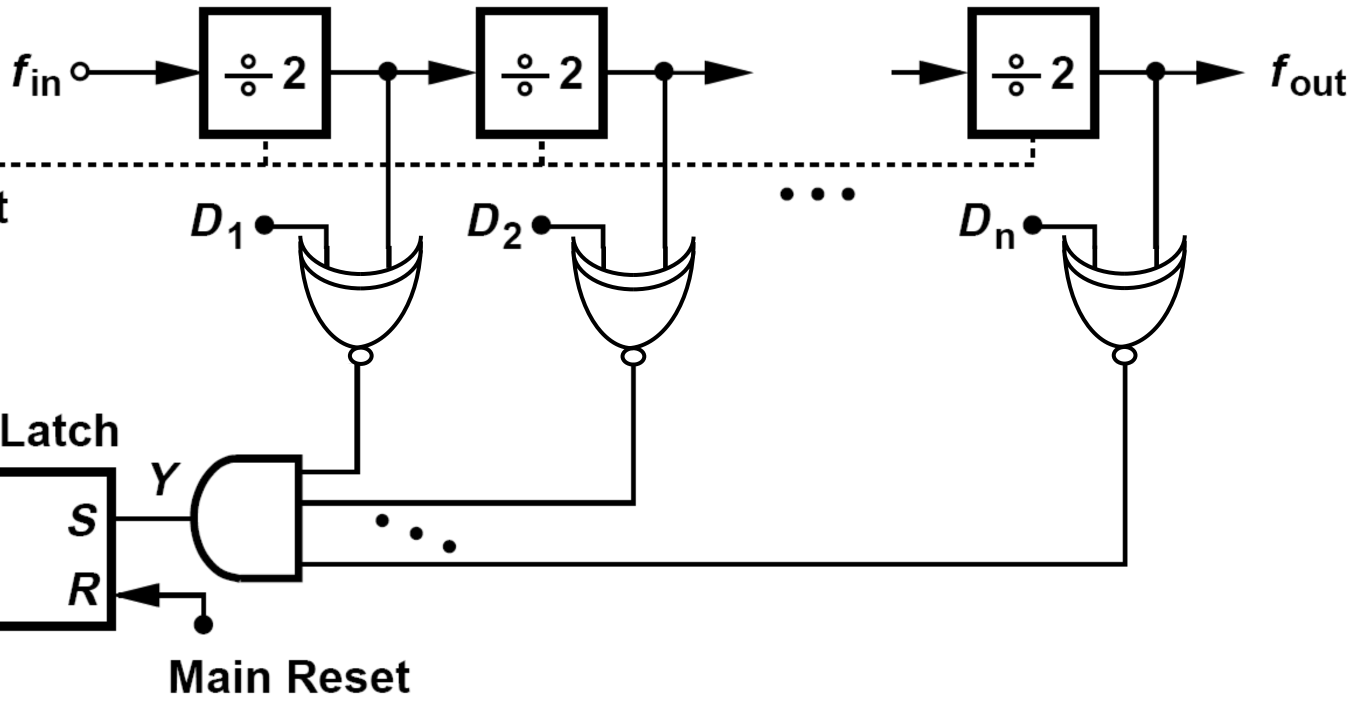


- LC cross coupled pair oscillator as the mixer
- V_{out} as mixer LO port, V_{in} as mixer RF input port
- Input lock range $\Delta\omega_{in} = \frac{\omega_0}{Q}\left(\frac{4}{\pi}\frac{I_{in}}{I_{osc}}\right)$. Higher injection current leads to wider locking range.
- Tradeoff between lock range and phase noise due to tank Q.

Ref [1]

Counter Implementation

Ref [1]



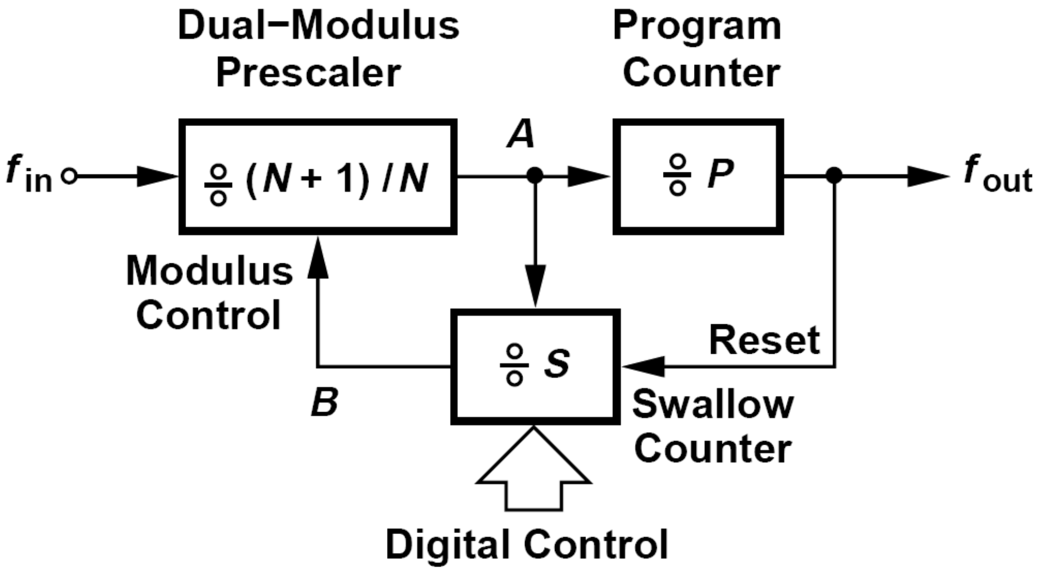
- Asynchronous counter
  - D_1 ~ D_n set the counter value
  - XNOR gates compare div2 output counts with the present D1~D2.
  - The counter is reset by the RS latch once the counter output reaches the present value.

☺ Easy to design, flexible programmability, low power

☹ Reset is generated from the output of all stages. Propagation delay from the first to the last stage limits the maximum speed.

Pulse Swallow Divider

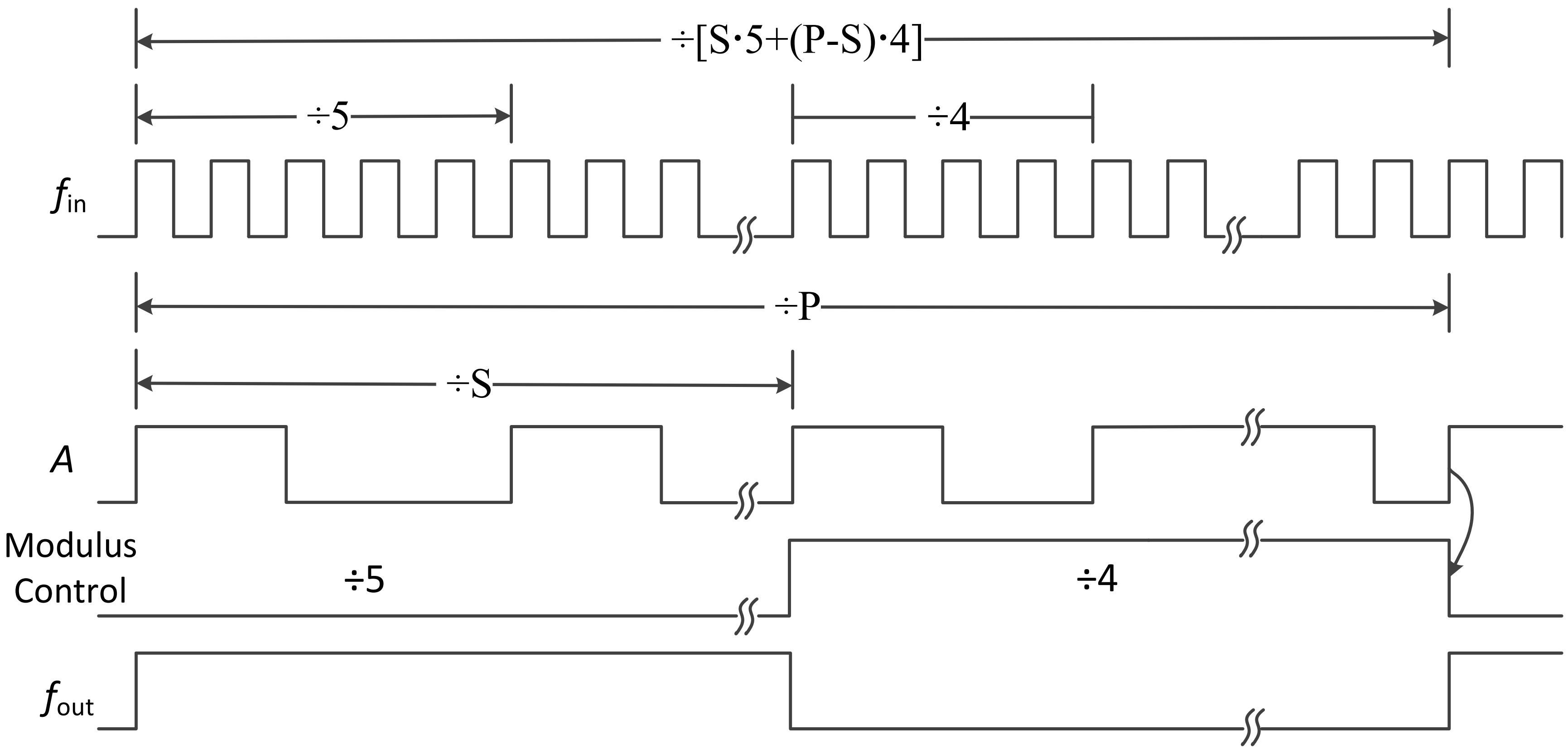
- In practice, *N* ranges from a single digit to a few hundred or more.
- N divider input clock from VCO can be as high as few GHz or more.
- Challenging to design a divider with large and programmable divide ratio operating at high VCO frequency → Pulse swallow divider
- A prescaler with only two small divide ratios divides high input frequency down first, followed by two counters with large divide ratios operating at lower frequency.



$$\frac{f_{out}}{f_{in}} = N \cdot P + S$$

Ref [1]

Pulse Swallow Divider



- Three dividers
  - Dual modulus prescaler ÷*N*/*N*+1 (*N*=4 in the figure)
  - Program counter ÷*P*
  - Swallow counter ÷*S*

1. Reset all counters
2. Prescaler first divides by *N*+1
3. Swallow counter counts to S and set prescaler divide ratio to *N*.
4. Program counter counts the output pulses from prescaler to *P* and resets all counters.

- During one output period, prescaler counts by N+1 for S times and by N for (P-S) times ➔Total divide ratio is
  $(N+1)\cdot S+N\cdot(P-S)=N\cdot P+S$
- P > S !

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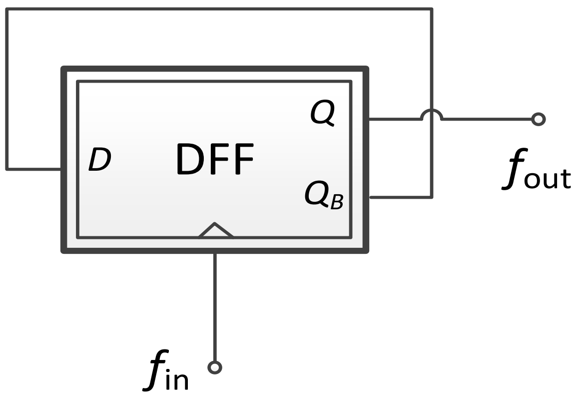
Prescaler – Dual Modulus Divider

Why is a prescaler needed?

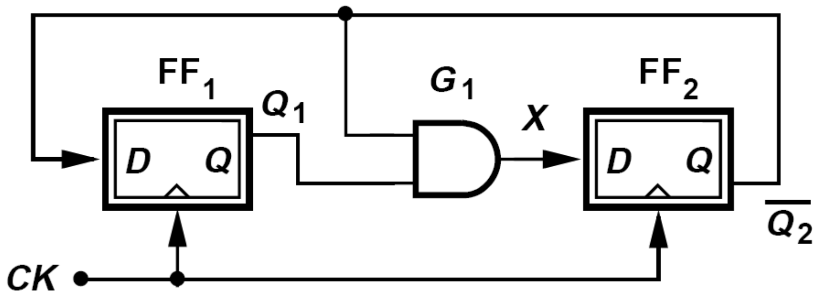
Challenging to design a divider with large and programmable divide ratio operating at high VCO frequency

➔ Pulse swallow divider with prescaler as the first stage



- Simple Div-2 divider
- The state machine has 2 states

D (QB)	Q
0	1
1	0



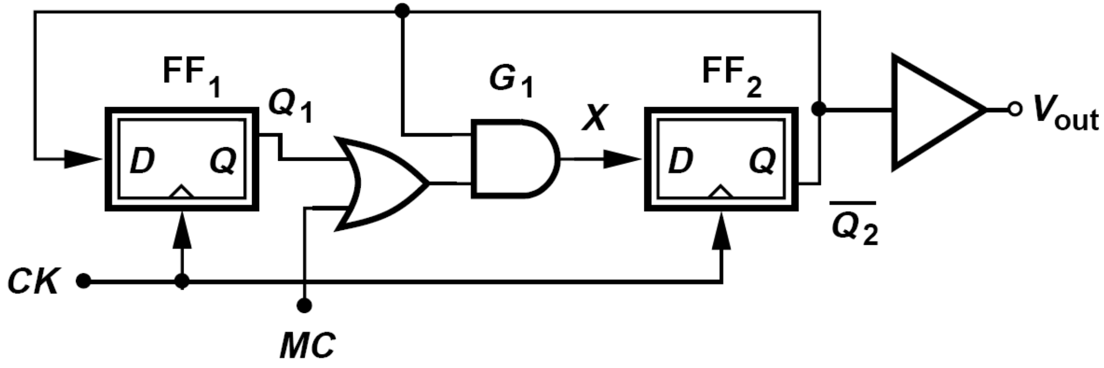
- Div-3 divider
- The state machine has 3 states
- If initial state is 00, it goes to 01 at the next state
- If initial state is one of the 3 counter states, it will loop within these states.

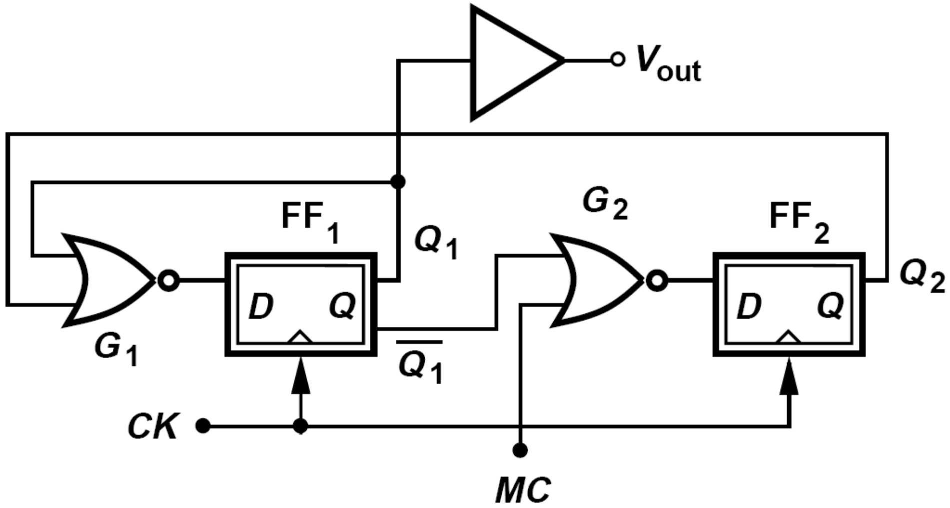
Q1	Q2B	X
0	0	0
0	1	0
1	1	1
1	0	0

- To implement Div-N, required DFF count is $\lceil \log_2 N \rceil$

Prescaler – Dual Modulus Divider

Div-2/3 Prescaler





- MC: modulus control signal from S counter
- MC=0: Div-3
- MC=1: Div-2; DFF1 is not part of the loop, DFF2 with Q_{2B} feedback to D_2

- Same Div-2/3 functionality
- Each DFF output drives only logic gates, not another DFF input ➔ smaller loading, higher speed

Ref [1]

Considerations in Divider Topology

- Logic dividers (N divider in PLL)
  - Input (from VCO) differential or single-ended
  - Input swing
  - Loading to VCO (divider input capacitance)
  - Maximum speed
  - Minimum speed (for dynamic divider)
  - Output swing
- LO dividers (driving Rx or Tx mixers)
  - Differential, I/Q 4-phases, or multi-phase
  - Duty cycle
  - Input swing
  - Output swing and slew rate
  - Phase noise
  - Amplitude and phase mismatch (RSB)
  - Maximum and minimum speed

References

[1] Behzad Razavi, RF Microelectronics, Prentice Hall, 2nd edition, 2012

[2] William F. Egan, Frequency Synthesis by Phase Lock, 2nd edition, 2000

[3] J. Yuan and C. Svensson, “High-speed CMOS circuit technique,” IEEE J. Solid-State Circuits, vol. 24, pp. 62–70, Feb. 1989

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